AL-NEELAIN UNIVERSITY

Faculty of Computer Science and Information Technology — Graduate Studies

Department of Computer Science | Artificial Intelligence Track

Course : COSC726 AGENTIC AI

مقترح بحث: نظام الوكلاء المتعددين للتشخيص التنبئي للسكري

Research Proposal: Agentic Multi-Agent Clinical Decision Support System for Diabetes Prediction with Multi-Provider LLMs and Quality Gates

[Academic Research Proposal]

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1. Executive Summary

Diabetes mellitus is among the fastest-growing global chronic health challenges, requiring timely, personalized, and explainable clinical interventions. Developed under Course : COSC726 AGENTIC AI at Al-Neelain University under the academic supervision of Dr. Fakhreldeen Saeed, this research project advances traditional diagnostic systems by evolving from isolated, black-box predictive models into an Agentic Multi-Agent Clinical Decision Support System (CDSS) powered by Multi-Provider Large Language Models (LLMs: Ollama, Cohere, DeepSeek, Gemini, OpenAI) and an offline deterministic expert engine, strictly governed by Clinical Quality Gates. The system simulates a multidisciplinary medical team comprising five specialized AI agents: a Clinical Triage and Biological Validation Agent, a Calibrated ML Predictive Analytics Agent, a Clinical Reasoning Explainer Agent, a Personalized Lifestyle and Nutrition Agent, and a Safety and Governance Gatekeeper. Enforcing biological boundaries, gender-specific checks, ADA guideline compliance, and real-time emergency triage safeguards, the solution pairs reliable statistical accuracy with clinical explainability and safe patient guidance.

2. Problem Statement & Clinical Rationale

Traditional machine learning applications for diabetes prediction encounter critical limitations that hinder clinical deployment:

1. The 'Black-Box' Interpretability Void: Conventional classifiers yield raw probability scores without physiological reasoning, leaving patients and physicians without context.
2. Absence of Biological Integrity & Gender Checks: Legacy predictors frequently accept biologically implausible inputs (e.g., Blood Glucose = 0, or pregnancies recorded for male patients) without validation.
3. Lack of Actionable Lifestyle Prescriptions: Standard models stop at numerical prediction and fail to provide personalized dietary or physical exercise regimens.

4. Absence of Unified Biomarker Importance Visuals: Interfaces lack interactive, multi-dimensional feature importance visualizers for clinician exploration.
5. Deficit in Medical Safety and Regulatory Governance: Standard systems lack protective guardrails to detect acute diabetic crises (Glucose >= 250 mg/dL) and enforce clinical disclaimers.

> **[Research Significance]**
> This project bridges the translational gap between machine learning algorithms and bedside clinical safety by establishing an agentic, explainable, and multi-gated architecture aligned with American Diabetes Association (ADA) standards.

3. Research Objectives

The key research and development objectives are structured as follows:

#	Objective	Scope & Technical Implementation
1	Calibrated Clinical Modeling	Develop and calibrate ensemble ML algorithms (CalibratedClassifierCV) on preprocessed biomarker datasets, ensuring reliable risk probabilities and feature rankings.
2	Multi-Agent System Architecture	Architect 5 specialized autonomous agents communicating through strict Pydantic data contracts to orchestrate comprehensive patient triage.
3	Multi-Provider LLM Integration	Integrate 5 LLM providers (Ollama, Cohere, DeepSeek, Gemini, OpenAI) and an offline deterministic ADA expert engine for robust reasoning.
4	Implementation of Quality Gates	Enforce nine rigorous verification gates across all phases addressing biological validity, model calibration, semantic alignment, and emergency safety.
5	Interactive Web Platform & Criteria Tab	Construct a responsive Streamlit dashboard featuring Arabic (RTL) and English support, pipeline tracking, criteria analysis tab, and shape switcher.
6	Sanitized Clinical Document Automation	Generate verifiable patient diagnostic dossiers with sanitized headers and academic phase reports in Microsoft Word (.docx) format.

4. Proposed Multi-Agent System Architecture

The system models an interactive Multidisciplinary Clinical Team (MDT) operating in a sequential pipeline with intermediate quality gates:

Specialized Agent	Clinical Responsibilities & Technical Gate
Clinical Triage & Validation Agent	Sanitizes raw biomarkers, detects anomalous zero-values, checks gender logic (disabling pregnancy for males), and assigns priority tiers.
Predictive Analytics Agent	Executes the calibrated ML pipeline, derives true posterior probabilities, and computes biomarker risk factor contributions.
Clinical Reasoning & LLM Explainer	Translates metrics into comprehensive clinical reasoning aligned with ADA clinical standards in both English and Arabic across multi-model providers.
Lifestyle & Nutrition Agent	Formulates tailored dietary modifications (glycemic index control), BMI-adjusted exercise routines, and lab follow-up schedules.
Safety & Governance Gatekeeper	Conducts final quality audits, enforces legal medical disclaimers, intercepts emergency crises (>= 250 mg/dL), and prevents hallucinations.

5. Quality & Clinical Safety Gates

[Enforced Quality Gates]

- Gate 1 (Biological Integrity Gate): Rejects and flags biologically unviable records (e.g. fatal glucose values) before prediction and validates gender constraints.
- Gate 2 (Model Performance & Calibration Gate): Mandates ROC-AUC >= 0.80 and validated Brier calibration score with sub-100ms inference.
- Gate 3 (Semantic Alignment Gate): Guarantees zero contradiction between numerical statistical risk and LLM narrative, enforcing standard clinical wording.
- Gate 4 (Lifestyle Feasibility Gate): Checks patient contraindications (e.g., prohibiting high-intensity cardio for severe BMI/elderly).
- Gate 5 (Governance & Emergency Triage Gate): Enforces critical red-flag triggers for blood glucose exceeding 250 mg/dL, prompting immediate emergency consultation.

6. Implementation Phases & Deliverables

Phase	Dedicated Directory	Key Milestone & Deliverables
Phase 1: Data Modeling & Validation	Phase_1_Data_Modeling_and_Validation	Imputation of zero-biomarkers, calibrated ensemble modeling, and Gate 1-2 verification.
Phase 2: Multi-Agent Architecture	Phase_2_Multi_Agent_Architecture	Pydantic contract schemas, 5-agent sequential orchestration, multi-provider LLM engine with ADA fallback.
Phase 3: Interactive UI & Integration	Phase_3_Interactive_UI_and_System_Integration	Streamlit dual-language application, criteria analysis tab, shape switcher, and patient Word report generation.
Phase 4: Verification & Benchmarking	Phase_4_Verification_Testing_and_Benchmarking	Stress testing on extreme/outlier patient cases, automated test suites (10/10), and latency benchmarks.

7. Academic & Clinical References

1. American Diabetes Association (ADA). Standards of Care in Diabetes—2024 / 2026. Diabetes Care.
2. World Health Organization (WHO). Global report on diabetes. Geneva: World Health Organization.